האוניברסיטה הפתוחה
THE OPEN UNIVERSITY OF ISRAEL
الجامعة المفتوحة

Identifying Likely Large-Scale Agile Implementation Pathways

Erez Morabia

Supervisor: Prof. Shmuel Tyszberowicz

M.Sc. Advanced Project in Computer Science · 22997

The challenge: choosing the next practice

Which practices should this team focus on next?

35
practices

Different starting points The next practice should reflect the team's current maturity profile.

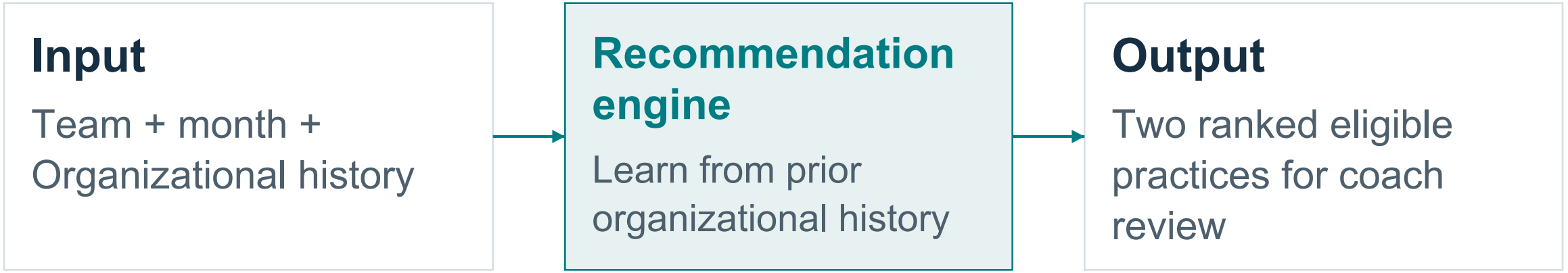
Relevant experience Other teams' improvement histories provide evidence about possible next steps.

Difficult comparison Manually comparing teams and histories becomes difficult at organizational scale.

The challenge: turn organizational experience into two team-specific practices for coach review.

Learning organizational improvement behavior

Use observed team histories to identify likely next practices.



Component-specific scoring windows; historical learning has no fixed lookback.

The contribution: empirical organizational learning

The ranking supports coach review. Scores express relative evidence.

Three signals form one ranking

Collaborative filtering

Match maturity profiles and learn from peer improvements

Practice Transition Model

Observed practices following the team's recent improvements

Time-aware popularity

Count improvements across teams: earlier history + latest interval

Normalize components and blend their scores

Weights come from completed earlier prediction-month outcomes.

Rank eligible practices and return two recommendations

Popularity balance uses completed earlier outcomes. Eligibility excludes practices at maximum maturity.

Project report Executive summary, pp. 5-6; §3.3-3.5

Representing the organizational history

87
teams

35
practices

10
recorded months

654
unique team-month snapshots

One snapshot = one team's maturity scores on one date.

One prediction case uses the latest earlier assessment + up to 3 later snapshots.

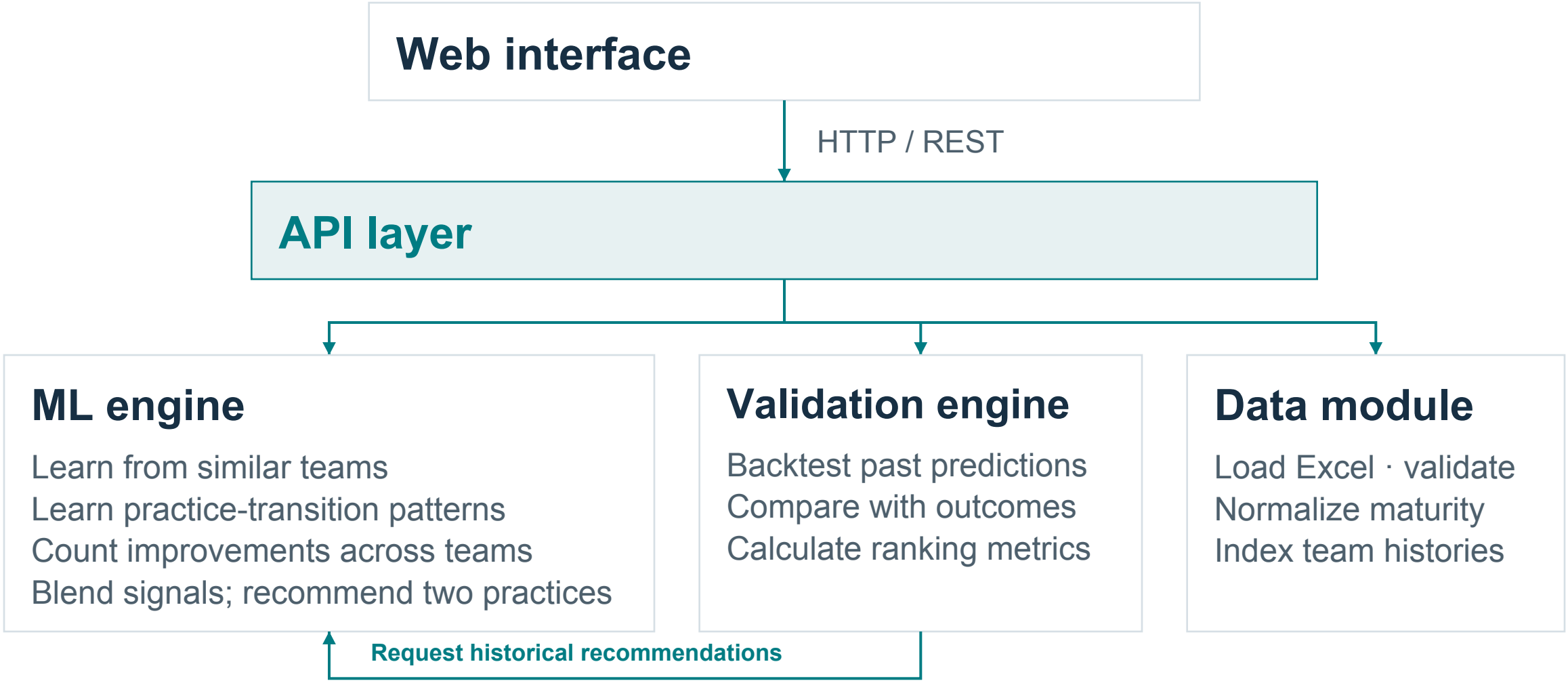
Jan → Mar → Apr
Initial history

May → Jun → Jul → Aug → Sep
Primary prediction months

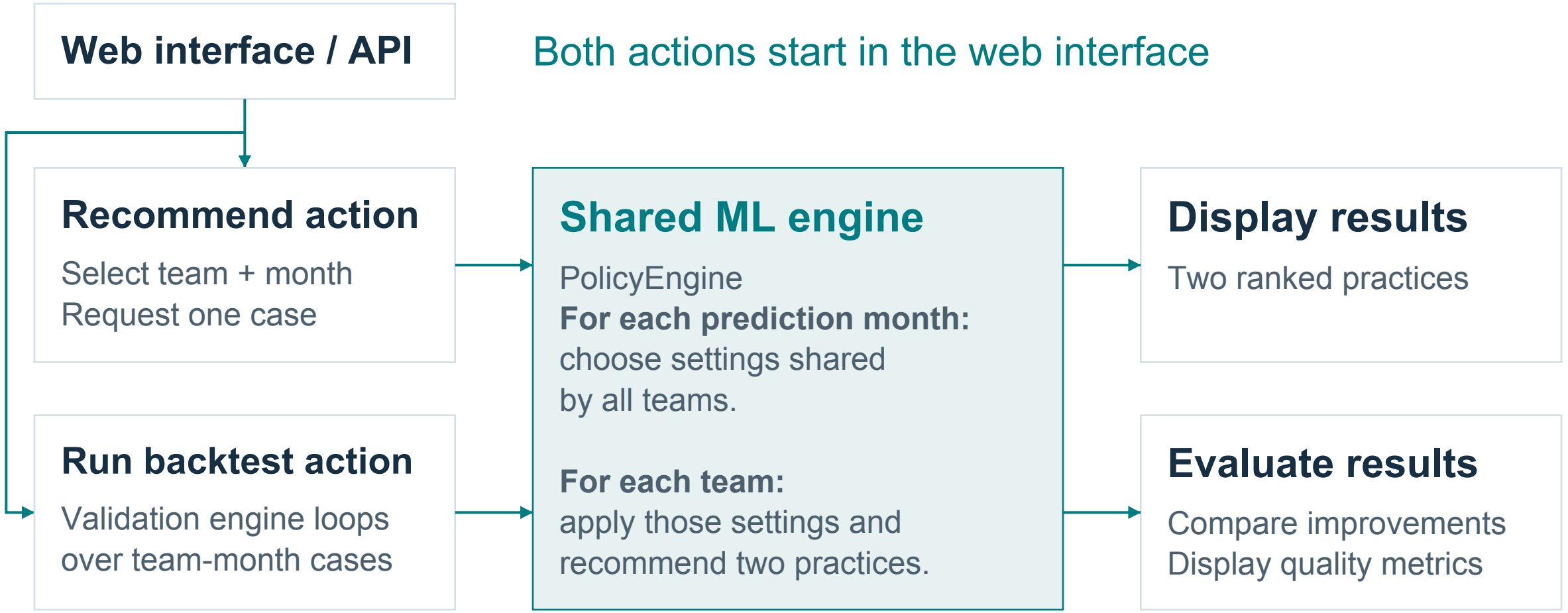
Oct → Nov
Follow-up

Reserve the first 3 recorded snapshots for initial history.
First evaluated prediction: May; latest earlier assessment (baseline): April.

High-level software architecture



One shared engine serves the application



One case for Recommend; many historical cases for backtesting. Same ML engine.

One recommendation request, end to end

Example · Team: Black Pearl · Prediction month: August 2020

Which two practices should this team prioritize next?

Find current state

Latest assessment:
5 July 2020
28 eligible practices
below maturity level 3

→

Apply policy

Top 10 of 47 qualifying peer teams, matched by historical profile:
cosine similarity ≥ 0.75

→

Score and rank

Normalize and blend the three signals;
rank all 28 candidates

August 2020 policy

Similarity 25% · Transitions 25% · Popularity 50% (historical only)

Each component uses its own history

Black Pearl · August prediction · baseline: 5 July 2020

Component	History used for this recommendation
Similarity	Find earlier peer assessments similar to Black Pearl's July state. Check improvements in their next two assessments. Use no peer observations after July.
Practice transitions	Learn practice-following patterns from January-June history. Compare Black Pearl's July scores with its May and June scores. Collect improved practices once, then score practices that historically followed them.
Popularity	Count improvements across teams in January-June. August's policy uses historical popularity only.

Monthly policy: one shared set of settings for all teams.
For August, test combinations across May prediction cases using their May-July outcomes.

Source: src/ml/policy.py and src/ml/sequences.py; project report §3.3-3.7

Why these two practices rank highest

Black Pearl · August 2020 · 28 eligible practices ranked

Each component: raw evidence ÷ highest raw value. Final rank: weighted total.

Component / weight	#1 Scrum Master Selected	#2 Tech debt strategy Selected	#3 DoD Ranked third
Similarity · 25%	**0.000** (0 ÷ 0.5944)	**0.513** (0.3047 ÷ 0.5944)	**1.000** (0.5944 ÷ 0.5944)
Transitions · 25%	**1.000** (1.333 ÷ 1.333)	**0.525** (0.700 ÷ 1.333)	**0.000** (0 ÷ 1.333)
Popularity · 50%	**1.000** (15 ÷ 15)	**0.933** (14 ÷ 15)	**0.800** (12 ÷ 15)
Weighted total	**0.750**	**0.726**	**0.650**

Similarity: combine how similar each peer was with how much it improved the practice.
Transitions: combine historical evidence that a practice follows Black Pearl's recent improvements.
Popularity: count how often the practice improved across teams.
Each component is scaled relative to its highest raw score. Details: backup A4.

Temporal controls prevent future-data leakage

Example: July baseline, August prediction



Why three snapshots? Allow time for practice maturity to improve.

Select weights only after earlier outcomes are complete, to avoid future-data leakage.

May-July: popularity-only bootstrap.
August: May's three-snapshot outcomes are complete; policy selection can begin.

Three snapshots is a fixed design choice; we did not optimize the window length.

What counts as a successful ranking?

Conditional Hit Rate@2

At least one recommendation matches a later improvement
in a case where an improvement occurred.

Black Pearl Tech debt strategy later improved: this case is a hit.

298 eligible prediction cases with complete follow-up

May-September · baseline + 3 later snapshots · ≥2 eligible practices



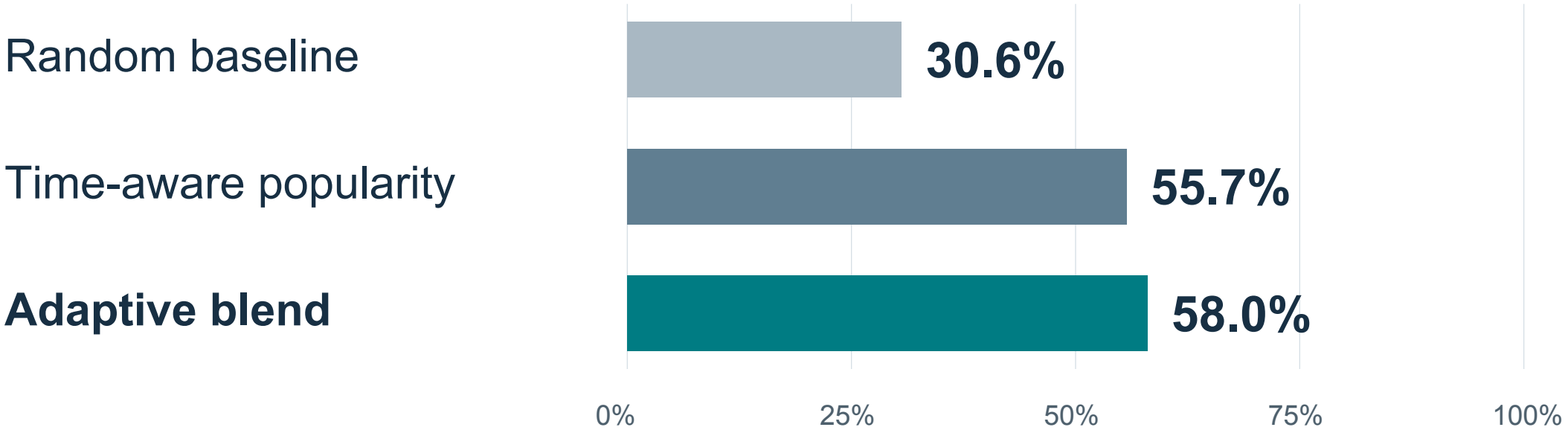
120 cases
Maturity increased

178 cases
No maturity increase recorded

Hit-rate cohort: 120 improvement cases + 1 with partial follow-up = 121.

Primary historical results

Mean monthly Conditional Hit Rate@2



Popularity counts how frequently practices improved across teams, with an adjustable emphasis on recent improvements.

1.9× random · +2.3 percentage points over popularity

Five primary months. 121 outcome-bearing cases. The difference is exploratory.

Capabilities, practical value and limits

Working decision aid

Excel input + web interface / API

Automatic monthly policy selection

Consistent, evidence-based rankings

Evidence boundaries

One organization, short history

3 of 5 months use bootstrap

Practical benefit needs a pilot

Ready for supervised pilot testing

The prototype still needs production hardening. Estimated time savings have not been measured.

Project report Executive summary, pp. 6-7; §7.3

A prospective pilot can test practical benefit

Recommendation-assisted coaching

Usual coaching

Compare improvement frequency and time to improvement.

Record actual attempts, failures and recommendation use.
Predefine allocation, duration and success criteria.

Conclusions

An empirical approach: learning from how teams improved provides evidence for what another team could consider next.

Implemented in a working recommendation system,
evaluated retrospectively, and ready for supervised pilot testing.

Questions and discussion

A1. Results by prediction month

2020	Cases	Blend	Popularity	Recommendation policy
3 May	21	28.6%	28.6%	Bootstrap / popularity only
8 June	22	63.6%	63.6%	Bootstrap / popularity only
5 July	27	66.7%	66.7%	Bootstrap / popularity only
3 August	24	79.2%	75.0%	Adaptive blend
6 September	27	51.9%	44.4%	Adaptive blend
5 October	24	33.3%	20.8%	Adaptive blend*
4 November	6	33.3%	33.3%	Adaptive blend*

Bootstrap: no earlier prediction month has complete three-snapshot follow-up to select blend weights.
Uses 100% popularity (50% historical + 50% recent); similarity and transitions have zero weight.
Primary: 58.0% vs 55.7%. All seven months: 50.9% vs 47.5%.
*October and November: incomplete follow-up; supplementary results, excluded from the primary average.

A2. Policy search grid

Select one shared policy per prediction month, using only completed outcomes from earlier predictions.

Parameter	Candidate values	Meaning
Peer count	5, 10 or 19	Maximum number of distinct peer teams used for similarity evidence.
Minimum cosine similarity	0, 0.50 or 0.75	Keep peers at or above this threshold; higher values require closer profiles.
Three component weights	15 combinations in 25% steps	Similarity + transitions + popularity = 100%. Example: 25% / 25% / 50%.
Recent share within popularity	0%, 25%, 50%, 75% or 100%	0% = historical only; 100% = recent only. Intermediate values mix the two scores.

3 × 3 × 15 × 5 = 675 policies. Select the highest mean monthly hit rate on completed earlier outcomes.

A3. One shared policy is selected each month

1. Gather evidence Use earlier prediction months whose three-snapshot outcome windows have closed before the target month.

2. Compare policies Replay all 675 combinations. Select the highest mean monthly Conditional Hit Rate@2 on that earlier evidence.

3. Apply the policy Use one set of settings for all teams in the target month. Each team's evidence still produces its own ranking.

Prediction month	Earlier predictions used	Their outcome windows
August	May	May-July
September	May and June	May-July and June-August

No completed earlier outcomes: use default popularity. Reselect each month; settings may stay the same.

A4. How component scores are normalized

Tech debt strategy · Black Pearl · August 2020

Component	Tech debt strategy raw score	Highest raw score and practice	Division and result
Similarity	0.3047	0.5944 (DoD)	0.3047 ÷ 0.5944 = **0.513**
Transitions	0.700	1.333 (Scrum Master)	0.700 ÷ 1.333 = **0.525**
Popularity	14 events	15 events (Scrum Master)	14 ÷ 15 = **0.933**

Normalized score = raw evidence ÷ largest raw value for that component.

Similarity: sum of peer similarity × observed maturity increase.
Transitions: sum of learned successor probabilities for recent improvements.
Popularity counts improvement events across teams. Values shown are rounded; full precision is used.

Source: src/ml/policy.py; source workbook; project report §3.5 and §3.7

A5. Where raw scores and divisors come from

Black Pearl · August 2020 · the two inputs to each division in A4

Component	Tech debt strategy's raw score	Highest raw score (divisor)
Similarity	ACM: 0.913998 × (1 ÷ 3) = **0.3047**	DoD: ACM + Puffin 0.3047 + 0.2897 = **0.5944**
Transitions	Product Owner + Scrum Master 0.500 + 0.200 = **0.700**	Scrum Master: Backlog grooming + Retro 2/6 + 1/1 = **1.333**
Popularity	Tech debt strategy: **14 improvement events**	Scrum Master: **15 improvement events** (maximum)

Similarity: ACM improved Tech debt strategy from level 1 to 2; the other nine selected peers contributed zero.
For DoD, ACM and Puffin each improved one level. Each contribution is peer similarity × (level increase ÷ 3).
Transitions: values are learned probabilities that the candidate follows Black Pearl's recently improved practices.
Scrum Master followed Backlog grooming in 2 of 6 successor links, and Retro in 1 of 1. Retro's evidence is one observation.
Popularity counts events across teams before July. Values shown are rounded; calculations use full precision.

Source: source workbook; src/ml/policy.py and src/ml/sequences.py

A6. Code walkthrough

1. _compute_components

src/ml/policy.py:335

Collect eligible practices and raw evidence for the three signals.
Inputs: team + latest assessment date. Output: CaseComponents.

2. score_case

src/ml/policy.py:544 · inputs: CaseComponents + selected policy

Select peers, normalize evidence, mix historical/recent popularity,
and apply policy weights. Return a score for each eligible practice.

top_practices() then selects the two highest-scoring practices.

Source: src/ml/policy.py · PolicyEngine._compute_components and score_case